

Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
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Review / Exercises



Ex 1: Suppose investors can only invest in the following two assets. All investors are risk averse:

<u>Stocks</u>	<u>E(R)</u>	<u>Std.Dev.</u>
ABC	\$45	20%
XYZ	\$32	10%

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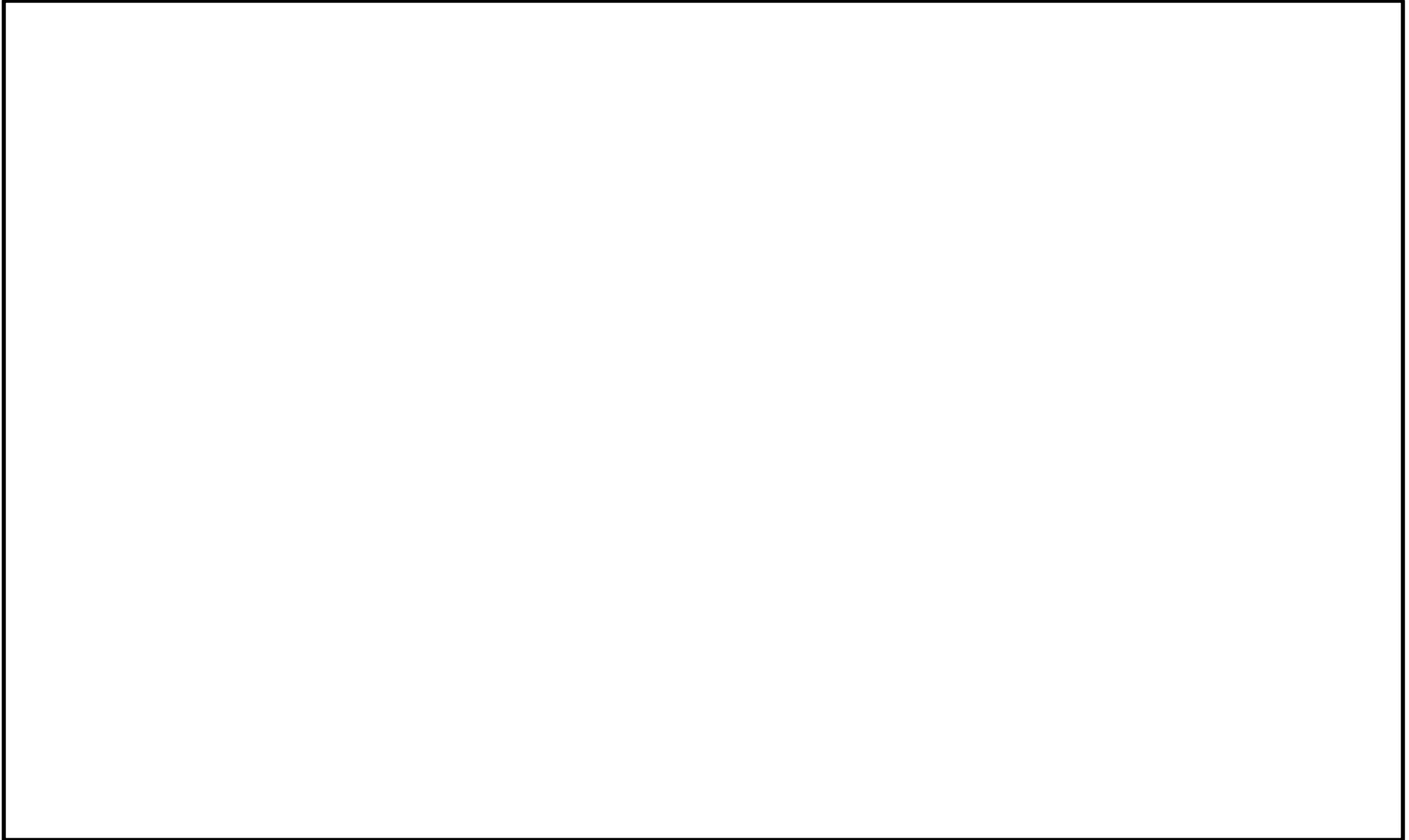
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Q2: What happens if the std. dev of XYZ increases by 15 pct. points (from 10% to 25%)? Does the investor prefer stock ABC or XYZ?

A2: The demand for XYZ goes down. Actually, the demand for XYZ goes to zero: ABC has same risk but higher return, so, investor will only buy ABC stocks

Ex 2: What is the YTM on a zero-coupon bond with the face value of \$1,500, with 2 years left until maturity, and selling for \$1,200?



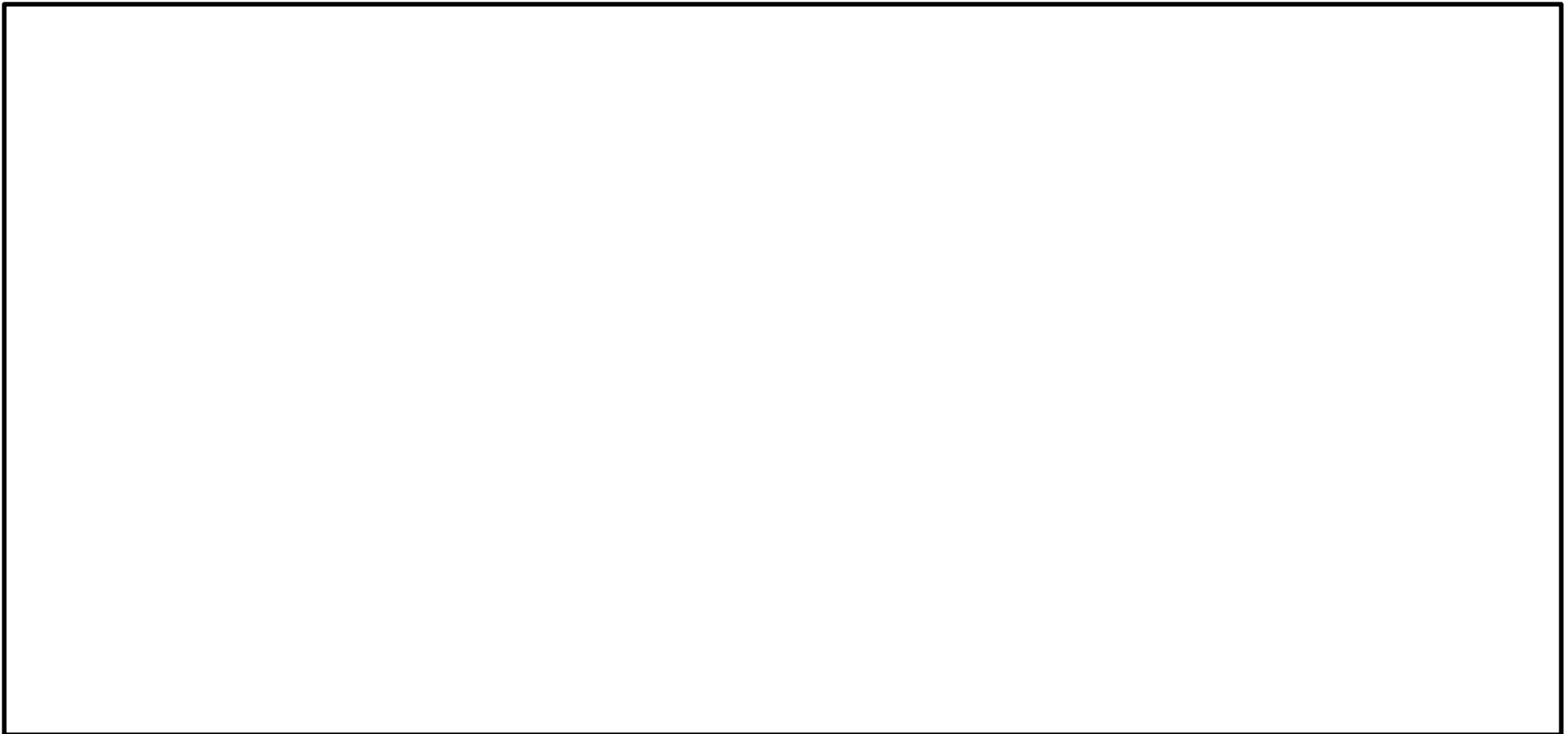
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A2: $\$1,200 = \frac{\$150}{(1+i)} + \frac{\$150 + \$1,500}{(1+i)^2}$

Trick:

1. Define $x = (1 + i)$; equation becomes: $1,200x^2 - 150x - \$1,650 = 0$

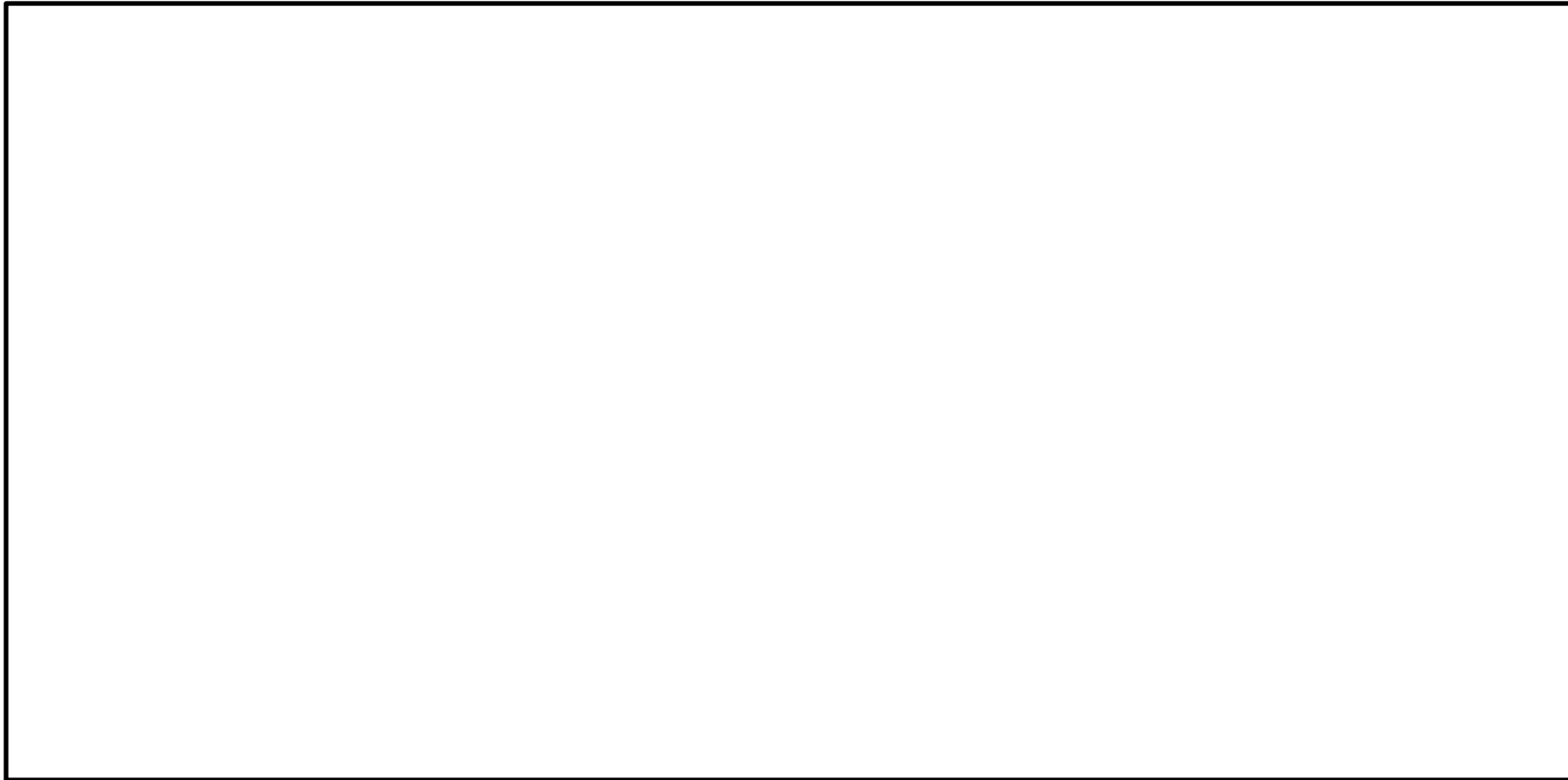
2. Divide by 150 and obtain: $8x^2 - x - 11 = 0$

3. Solve: $x = \frac{1 \pm \sqrt{1+352}}{16}$

4. Keep only the positive root; so $x = 1.2368 \Rightarrow i = 23.68\%$

Note that you have to use coupon rate ($i_{cr} = C/FV$) and PV, not directly current yield ($i_{cy} = C/P$), because from Lecture 2 the current yield “is a useful approximation to the YTM for long-term bonds, with price near par.” Here: only 2 periods, and bond is below par value ($P < FV$).

Q: What if it paid coupon semi-annually?



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A:
$$P = \frac{\frac{C}{2}}{\left(1+\frac{i}{2}\right)} + \frac{\frac{C}{2}}{\left(1+\frac{i}{2}\right)^2} + \frac{\frac{C}{2}}{\left(1+\frac{i}{2}\right)^3} + \left[\frac{\frac{C}{2}}{\left(1+\frac{i}{2}\right)^4} + \frac{FV}{\left(1+\frac{i}{2}\right)^4} \right]$$

$$\$1,200 = \frac{\$75}{\left(1 + \frac{i}{2}\right)} + \frac{\$75}{\left(1 + \frac{i}{2}\right)^2} + \frac{\$75}{\left(1 + \frac{i}{2}\right)^3} + \frac{\$75 + \$1,500}{\left(1 + \frac{i}{2}\right)^4}$$

$$i \approx 23.04\%$$

Need a computer to solve it (*i.e.* you won't be asked in an exam to provide a number)

Ex 3: A firm just paid an annual dividend of \$5 per share. You estimate that the dividend will grow at an annual rate of 2% from this year to the next and then at 4% forever.

Q: If the fair stock price is \$125 today, what must be the required rate of return?

A:

$$P_0 = \frac{D_1}{1 + k_e} + \frac{P_1}{1 + k_e} \Rightarrow P_0 = \frac{D_1}{1 + k_e} + \frac{1}{1 + k_e} \left[\frac{D_1 (1 + g_1)}{k_e - g_1} \right]$$

where $P_1 = \frac{D_1(1+g_1)}{k_e - g_1}$ (Gordon's formula)

Notice that it simplifies to (just this case):

$$P_0 = \frac{D_1}{1 + k_e} \left(1 + \frac{1 + g_1}{k_e - g_1} \right) = \frac{D_1}{k_e - g_1}$$

Plug in numbers, using that $D_1 = D_0(1 + g_0) = \$5 \cdot (1.02)$:

$$\$125 = \frac{\$5 \cdot (1.02)}{k_e - 0.04} \Rightarrow k_e - 0.04 = \frac{\$5 \cdot (1.02)}{\$125}$$

$$k_e = \frac{\$5 \cdot (1.02)}{\$125} + 4\% \Rightarrow k_e = 8.08\%$$

Ex 4:

Q1: What is the value of a call option on the expiration date if the strike price is \$300 and the spot price of the underlying asset is \$650?

Q2: What is the strike price of a put option that expires on the same day for the same asset and has a value of \$30?

Ex 4:

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Q2: What is the strike price of a put option that expires on the same day for the same asset and has a value of \$30?

A2: $value_{long}^{Put} = \$30 = X - \$650 \Rightarrow X = \$680$

Ex 5: Two firms, XYZ and ABC, are similar in every respect. They issued bonds with the same face value and the same coupon rate.

Shortly after, XYZ's production falls unexpectedly. Firm ABC's production is unaffected.

The market expects XYZ to make all required bond payments with an 80% probability, with 20% probability debt holders receive nothing.

Q: is the price of XYZ lower or higher than ABC's? If the price of ABC's bond is \$900 , what is that XYZ's bond?

A: XYZ price should be lower to compensate for higher risk

If the price of ABC's bond is \$900, $P_{XYZ} = \$720$:

$$P_{XYZ} = 0.8 \times \$900 = \$720$$

Ex 6: A stock is selling today for \$55. A put option on this stock that expires today is selling for \$6.

Q: What must be the exercise price of this option?

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Q: What must be the exercise price of this option?

A: Option expires today, so make a zero profit if you exercise it:

$$\pi_{long}^{Put} = X - S - p = X - \$55 - \$6 = 0$$

Solving for X yields: $X = \$61$

Ex. 7: Early in 2024, a trader sold one future contract for September 30, 2024 delivery.

The contract was written for the delivery of one share. When the trader entered the market, the spot price was \$30 and the futures price for September 30 delivery was \$35.

Q1: What was the trader's gain/loss on this trade if he closed his position on September 23, 2024, when the spot price was \$54 and the futures price for September 30 delivery was \$66?

Q2: What was the trader's gain/loss on this trade if wait until September 30, when spot price is \$29?

A1: $t < T$ (Sep. 23 earlier than Sep 30)

Trader's profit :

$$\pi_t^{short} = F_0 - F_t = \$35 - \$66 = -\$31$$

They are in the margin account.

Note: we completely ignore the spot prices bcs $t < T$

A2: $t = T$ (Sep 30) and $S_T = \$29$

Trader's profit :

$$\pi_t^{short} = F_0 - S_T = \$35 - \$29 = \$6$$

They are in the margin account.

Note: that we completely ignore the rule that at delivery date $S_T = F_T$ because he/she did not trade the future before delivery, nor he sold the future at $t=T$.

Ex 8:

You have a stock indexed portfolio worth \$20 M today. The index's value today is \$1,000.

You enter a short position on 75 futures contracts, each with a price of \$1,200, for delivery in June 2026. The multiplier for this index futures contract is 100.

You close the position before delivery, when the index is \$900 and the futures price for June 2026 delivery is \$800.

Q: What is the net gain/loss on your entire portfolio?

A:

Calculate gain on futures and on spot price

Futures:

$$\pi^{short} = (\$1,200 - \$800) \times 100 \times 75 = \$3,000,000 = \$3M$$

Spot price:

$$\frac{\Delta S}{S} = \frac{\$900 - \$1,000}{\$1,000} = -10\%.$$

$$\text{So } \Delta V_{Port.} = \frac{\Delta S}{S} \cdot V_{Port.,0}$$

$$\Delta V_{Port.} = -10\% \cdot \$20M = -\$2M$$

$$\text{Alternatively, could say: } \#stocks = \frac{\$20M}{\$1,000} = 20,000$$

$$V_1 = stocks \cdot S_1 = 20,000 \cdot \$900 = \$18M$$

$$\Delta V_{Port.} = V_0 - V_1 = \$20M - \$18M = -\$2M$$

So **total gain** (portfolio + future) is:

$$\Delta V_{Tot} = \pi^{short} + \Delta V_{Port.} = \$3M - \$2M = \$1M$$

Ex 9.

Firms ABC and XYZ are in the same industry.

However, ABC has a much higher PE ratio than XYZ.

Q: Which of the following can explain this?

- I. ABC's earnings are expected to rise more quickly in the future than XYZ's earnings
- II. ABC is perceived by the market to be less risky than XYZ.
- III. Both XYZ and ABC are in a fast-growing industry.
- IV. ABC's current earnings per share are higher than XYZ's current earnings per share

- a) I and II only
- b) I, II and IV only
- c) II and IV only
- d) II, III and IV only

A: Fact: $\left(\frac{P}{E}\right)_{ABC,j} \gg \left(\frac{P}{E}\right)_{XYZ,j}$ where $j = \text{industry}$

- ✓ I. ABC's earnings are expected to rise more quickly in the future than XYZ's earnings: **TRUE**
- ✓ II. ABC is perceived by the market to be less risky than XYZ: **TRUE**
- ✓ III. Both XYZ and ABC are in a fast-growing industry: **FALSE**
 - *industry common trend should affect their PE ratios in the same way*
- ✓ IV. ABC's current earnings per share are higher than XYZ's current earnings per share: **FALSE**
 - $P/E \left(\frac{P}{E}\right)_{ABC,j} \gg \left(\frac{P}{E}\right)_{XYZ,j}$ is consistent with different *expected* earning, not *current* earning
- ✓ So, solution is a): I and II only

Ex 10.

A bond with a face value of \$ 1,000 and a coupon of 5% is selling for \$1,100. Which of the following is true?

Q: Which of the following is true?

- a) the yield to maturity is 5.5%
- b) the yield to maturity is 6%
- c) none of the others
- d) the yield to maturity is less than 5%

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A:

Recall property no. 3: $YTM > i_{coupon\ rate}$ if $P < FV$

Here opposite: $P > FV$, so $YTM < i_{coupon\ rate}$; then: $YTM < 5\%$

Solution: d)!